

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

2nd Semester of B. Pharm. Examination

University Theory Examination May, 2018

PH120 / PH128 PHARMACEUTICAL ENGINEERING-II

Date: 11/05/2018 , Friday

Time: 10:00 a.m. to 1:00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
 2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
 3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
 4. Figures to right indicate marks.
 5. Draw neat labeled sketches wherever necessary.
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SECTION – I

Q 1 Attempt all questions. Each question is of one mark. 20

1. The suspension to be filterd is known as _____
[A] Paste
[B] Slurry
[C] Emulsion
[D] Solution
2. Separatio of solid from liquid is known as Clarification when concentration of solid is _____.
[A] >10 %
[B]>15%
[C] >1 %
[D] <1 %
3. _____process of filtration is similar ti sieving.
[A] Straining
[B] Impingment
[C] Entanglement
[D] Attractive forces
4. Centrifugation process involve _____.
[A] Vnader waals force
[B] Repulsive force
[C] Attractive force
[D] Centrifugal force

5. Sedimentation centrifuge is working based on _____.
[A] Difference in densities of two phases
[B] Difference in viscosities of two phases
[C] Difference in valency of two phases
[D] Difference in fluidity of two phases
6. Short cycle automatic batch centrifuge is also known as _____.
[A] Semi continuous centrifuge
[B] Non perforated basket centrifuge
[C] Perforated basket centrifuge
[D] Continuous horizontal centrifuge
7. In distillation process feed liquid is known as _____.
[A] Distillate
[B] Condensate
[C] Distilland
[D] Distillatant
8. Recovery of solvent is possible by _____.
[A] Drying
[B] Evaporation
[C] Mixing
[D] Distillation
9. Rault's law shows relationship between _____.
[A] Density and temperature
[B] Temperature and viscosity
[C] Concentration and vapour pressure
[D] Concentration and viscosity
10. Evaporation is defined as removal of solvent from _____.
[A] Paste
[B] Slurry
[C] Solution
[D] Emulsion
11. Vapour pressure is _____ with evaporation
[A] Directly proportional
[B] Inversely proportional
[C] No effect
[D] Variable effect
12. Greater the surface area _____ will be evaporation.
[A] Lesser
[B] Greater
[C] No change

- [D] Variable effect
13. Drying of granuls will give _____ effect.
[A] Improvemnet in flow
[B] Decrease the flow
[C] No effect
[D] Variable effect
14. Non hygroscopic material contains _____ water
[A] Bound
[B] Unbound
[C] Free
[D] Does not contain water
15. Non porous and insoluble matetrial have Equilibrium moisture content _____.
[A] Near to '0'
[B] Near to '5'
[C] 5-15
[D] 2-4
16. Free moisture content = _____.
[A] Toal moisture content - Equilibrium moisture content
[B] Toal moisture content + Equilibrium moisture content
[C] Toal moisture content / Equilibrium moisture content
[D] Toal moisture content * Equilibrium moisture content
17. For drying of granules _____ dryer is used.
[A] Drum Dryer
[B] Roller dryer
[C] Film dryer
[D] Fluidized bed dryer
18. Deliquescent Substances have natural tendency to _____.
[A] Absorb water and liquiefy
[B] Only Absorb
[C] Loose water and liquiefy
[D] Only loose
19. Air and water is example of _____.
[A] Prinary refrigerant
[B] Secondary refrigerant
[C] Birefrigerant
[D] Coolant
20. As viscosity of liquid increases, rate of evaporation _____.
[A] No change
[B] Constant
[C] Decrease

SECTION – II

- Q 2** Attempt any **TWO** of the following
- A** Describe the mechanism of filtration. **05**
 - B** Discuss the classification of the centrifuge **05**
 - C** Write a note on perforated basket centrifuge. **05**
- Q 3** Attempt any **TWO** of the following
- A** Explain the various types of filtration. **05**
 - B** Describe conical disc centrifuge. **05**
 - C** Discuss the various factors influencing evaporation. **05**

SECTION – III

- Q 4** Attempt any **FOUR** of the following
- A** Write a note on plate and frame filter press. **05**
 - B** Describe horizontal tube evaporator. **05**
 - C** Write a note on falling film evaporator. **05**
 - D** Define distillation and write down various application of distillation. **05**
 - E** Discuss azeotropic and extractive distillation. **05**
 - F** Define following terms: **05**
 - (a) Drying (b) Moisture content (c) bound water (c) Unbound water
 - (d) EMC
- Q 5** Attempt any **FOUR** of the following
- A** Describe laboratory scale apparatus of distillation. **05**
 - B** Define Free Moisture Content (FMC) and discuss the FMC Curve in detail. **05**
 - C** Write a note on HVAC System. **05**
 - D** Discuss film drum dryer. **05**
 - E** Explain the theories of drying. **05**
 - F** Define dry bulb and wet bulb temperature. And discuss method to determine both. **05**
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